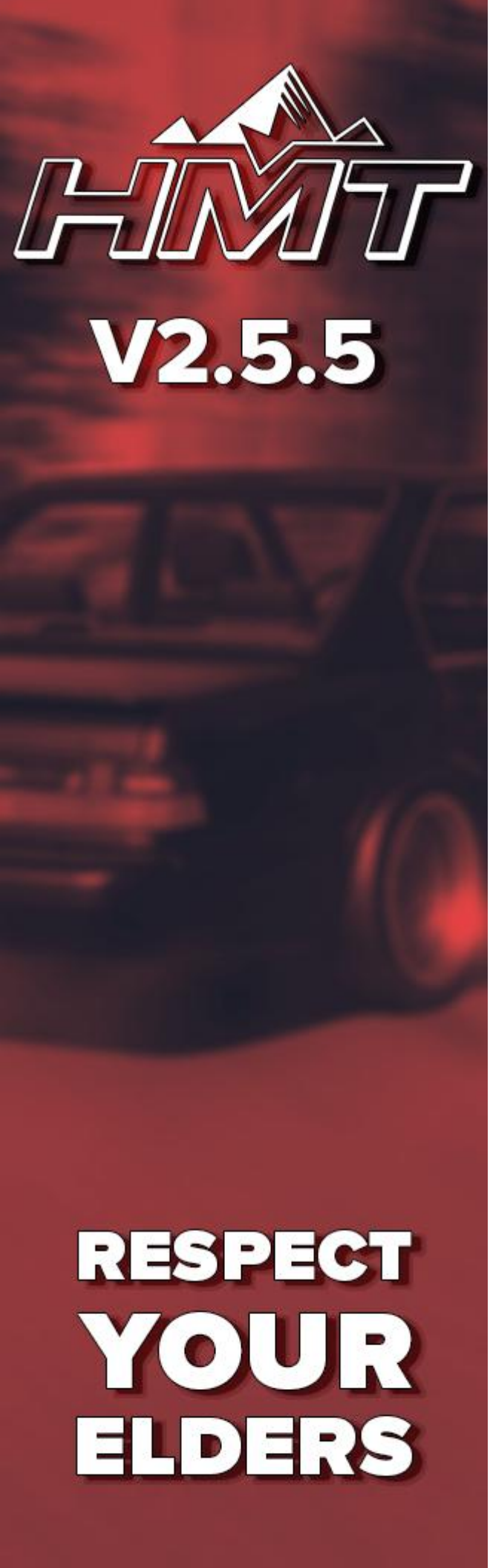


ETK I-Series Support pt.2

& Pure430b Gen 2 & Ablauf x HMT Aero

Harzer Mountain Tuning v2.5.5 *is a supplemental update adding and improving several products to the newest "Echo" product line for the I-Series. The strong reception of v2.5 has been incredibly encouraging, and this release reflects continued refinement across the I-Series platform, as well as a handful of additions to modern ETK platforms*!

The following pages detail the changes and additions included in this update.

About Harzer Mountain Tuning

Located in Halberstadt, Saxony-Anhalt on the northeastern edge of the Harz Mountains, ***Harzer Mountain Tuning*** *has spent over a decade building a reputation for performance parts found on some of the most finely tuned ETK vehicles on the road. Founded shortly after the launch of the current generation 800-Series,* ***HMT*** *quickly became the benchmark for both 800 and K-Series tuning.*

The ***Pure****,* ***Dusk****,* ***Eden****, and* ***Echo*** *packages represent a commitment not just to power, but to refinement. Backed by real R&D, dyno validation, and thousands of street miles on every platform we support, we don't cut corners, and we don't ask our customers to either.*

"Stärke und Präzision"

@XPOGOSAM

flickr

**Hyperlinks*



Misc. Things:

- **Added** HMT Euro plate
- Removed oldest TCU tune revision files
- Removed v0.27 Dusk ECU
    - Was a test file for something unimportant
- **Added** HMT Helical Differentials to front and rear slots of Modern ETKs
    - Same settings as our diff for the I-Series
- Returned Slate & Blackberry Interior Variants
- **Added** half cage to I-Series (WIP)
- Experimental Cold start LUA **added** temporarily
    - More info will come when system is out of Beta
- ***Reworked*** HMT Splash screen in modern cars with new logoset



NEW - HMT Modern Transmission Conversion Packages for I-Series:

To expand upon the popular 6-Speed Manual swap for the ETK I-Series, ***HMT*** now offers complete conversion packages for the modern ETK **8-Speed Automatic** and **7-Speed Dual-Clutch** Transmission.

Be it 6MT, 7-DCT, or 8AT, these swaps are **not** inexpensive, especially the new automatic options. But for those pursuing a no-expense-spared Echo build, they are options we made available.



6MT New Ratios

HMT has restructured the existing ratios available in the 6MT swap as well. In addition to the 3 OEM gearsets, there are now "HMT Remix" gear sets, which are a combination of different available standard ratios that result in either a close ratio or a long ratio gearbox. There is also a "Gt2 Tribute" gearset with very close ratios taken from a mid-2000s Gt2 class racecar.

Gear Display

All ***HMT*** transmission swap packages now include an optional period correct digital gear indicator to the right of the tachometer. It is fully glow-mapped to display all gear positions for both Manual and Automatic shifters, and displays current gear regardless, rather than just "D" in the automatics standard drive mode.

7 Speed Dual Clutch

Includes the same TCU Tune File slot we offer for the modern cars. This allows for use of our optimized ***HMT*** shift logic or an OEM+ Logic. Features **carbon** shifter bezel for differentiation.

DCT Gear Sets – (Patreon)

Two Gear Sets are available for the DCT Swap:

- Vanilla DCT Ratio
  - Same as F8X chassis BMWs
  - Designated as "Long Ratio"
- OEM DCT Ratios
  - Same as E9X chassis BMWs
  - Designated as "Close Ratio"

8-Speed Automatic

Dedicated 8AT TCU tune slot for all ***HMT*** supported platforms is in development, for now it can be tested in its **beta** form through the Pure Stage 1 Torque Converter, new from v2.5. Features **aluminum** shifter bezel for differentiation.

8AT Gear Sets – (Patreon)

Four Gear Sets are available for the 8AT Swap:

- Vanilla Ratios (Identical to BMWs 8HP70)
- OEM Ratios (Identical to 8HP75)
  - *Used by Alpina and others*
- OEM Ratios (Identical to 8HP51)
  - *Used by Toyota in A90 Supra*
- OEM Ratios (Identical to 8HP76)
  - *Used by BMW in G8X M Cars*



NEW - HMT Appearance Package for ETK I-Series:

HMT now offers period correct branding package for the I-Series!

These options were all designed to bring a subtle but distinct, period correct, livery to the I-Series all while allowing you to represent the HMT brand in a way that complements Echo program.

Whether it's a faithful retro-mod or a no-expense-spared build, these are a period correct touch that other enthusiasts will notice and immediately understand.



A special thank you to **@xyomos** *and* **@JumbledNick** *in my discord server for their help in developing and/or inspiring these assets.*



Decal Set

The ***HMT*** Decal Set is a tertiary colorable side stripe inspired by ***Hartge*** motorsport liveries of the period, bringing subtle old-school tuning house character to the I-Series bodywork. It's designed to complement both restrained and bold builds, and is available in four trim-compatible variants:

- **Window trim**: *Black* or *Silver*
- **Body trim:** *Black* or *Paint-Matched*

Banner/Sunstrip

The ***HMT*** Windshield Banner continues the period-correct motorsport branding for the I-Series, with a distinctive visual touch. ***Eight*** versions are available across ***two*** lettering styles and ***four***-color options:

Lettering: *Outline* or *Solid* lettering
Color: *White* or *Colorable* by all 3 color palettes





Remaster - HMT Badge and Logo Assets for Modern ETKs:

Remastered Textures

The signature ***HMT*** mountain logo has been fully remastered! Now reproduced from the ground up with higher resolution, and a more modular asset workspace (really only relevant for me). Every place the logo appears in this mod, from tailgate badges & trunk emblems to the navigation splash screen and V8 Engine cover badge, has been updated.

The new modularity makes special edition badges significantly easier to produce going forward. Including the original, there are now 5 badge editions available to all users: the Original, 300k Downloads, 400k Downloads, 500k Downloads, and “3 Years of HMT” celebrating this mod's birthday on April 19th.

A huge thank you again to **@xyomos**, a long-standing patron and member of the Discord server, whose massive contribution in remastering my assets for this made it possible.

Original	
300k Downloads	
400k Downloads	
500k Downloads	
3 Years of HMT	

There are also 2 additional versions available, a Bauhaus pattern version for Patrons, and a Memphis pattern for Boosters on my Discord

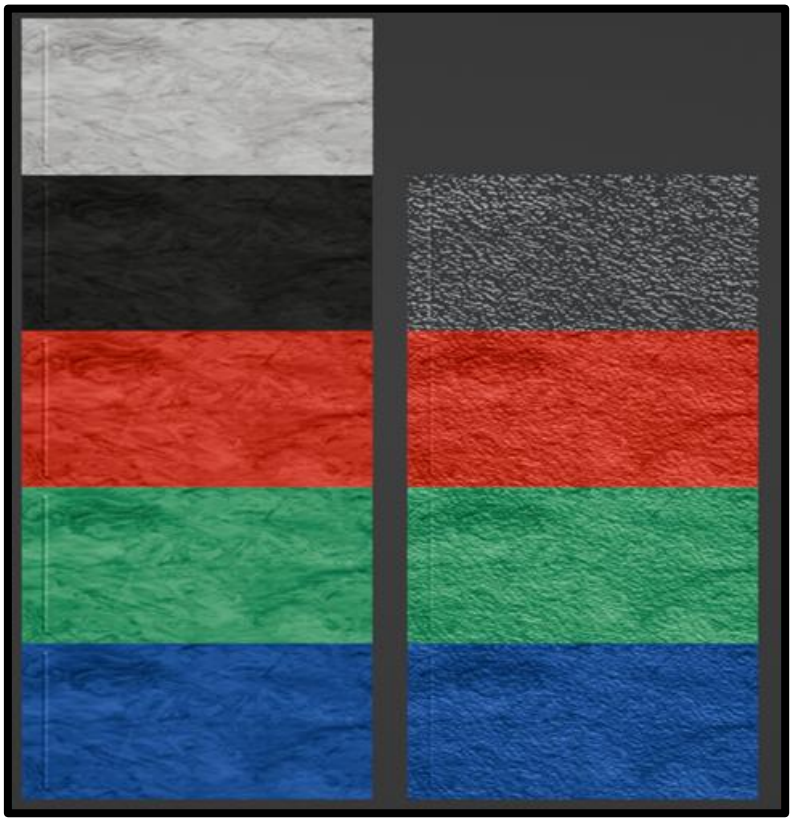






Changes/Additions - HMT Echo Turbocharger Package for I-Series:

Piping Retexture



The ***HMT*** Echo Kit piping has been fully retextured with 5 powdercoat color options across charge-pipes and intake pipe for both hybrid turbo and big turbo applications:

- **Gloss** ONLY
  - *Silver (Raw)*
- GLOSS OR WRINKLE
  - *Black*
  - *Colorable **Primary** Palette*
  - *Colorable **Secondary** Palette*
  - *Colorable **Tertiary** Palette*

Patrons also gain access to **Carbon piping**, in both carbon and carbon-Kevlar versions, bringing the total to **11 finish variants**.

**This system will find its way onto the modern vehicles in a light asset remaster coming soon*

New Turbo : Hollis H35-900



HMT now offers a 4th turbo in the Echo turbocharger ecosystem. The H35-900 from Hollis is the next step up from the H25-660. Benchmarked to its real-world equivalent, it takes everything from the H25-660 and scales it up, with the powerband shifting higher by a about ~2000rpm, with the tradeoff of much higher efficiency in the top end of the rev range.

We rarely advise it for street cars and find it most suitable for number focused builds or people who intend to race their I-Series in one form or another. We've seen output as high as 1050bhp from a fully built 3.4L that ***HMT*** put in a ttSport EVO.

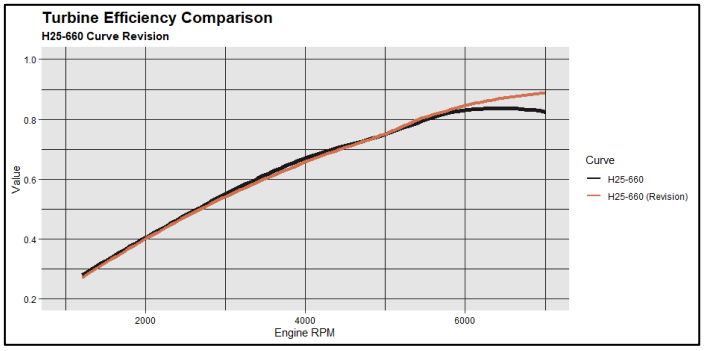
*Use H25-660 boost control settings, fully compatible.



Changes/Additions *Continued* - HMT Echo Turbocharger Package for I-Series:

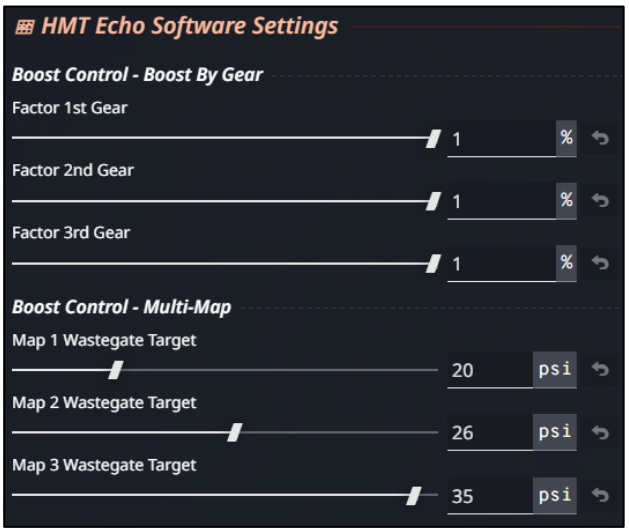
Mild Revision – H25-660

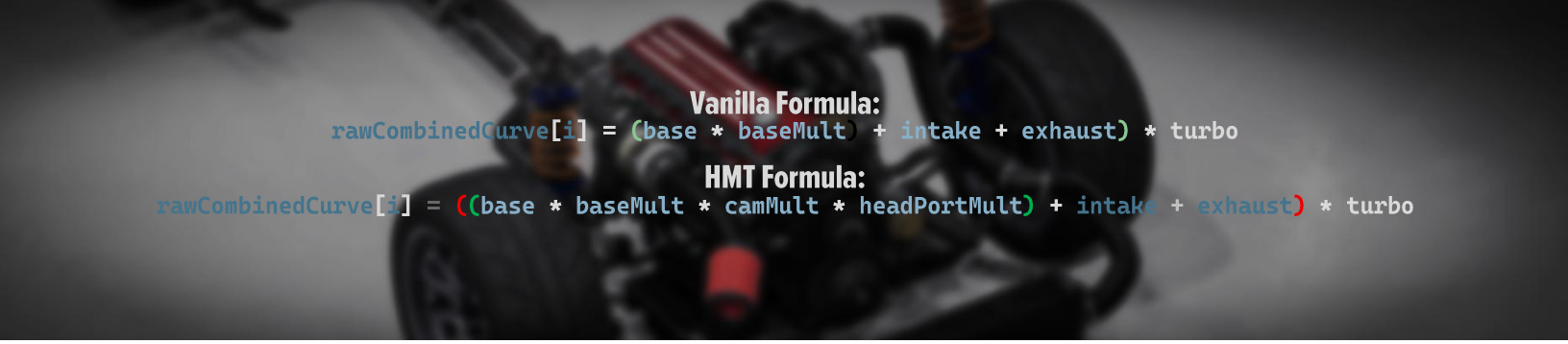
The Efficiency curve for the Hollis H25-660 has received some minor calibration. The top end efficiency is the largest change, as it is held further into the rev range now. Peak power remains the same but the tapering past peak has been reduced. Not visualized to the right is the changes to exhaust power, which are lightly reduced at mid-range RPM to keep spool speed realistic.



Added – Boost by Gear to Echo Standalone

Boost by Gear is now included in the ***HMT*** Echo standalone's boost control strategy as an additional option. Rather than setting a dedicated boost target per gear, it works as a simple multiplier, keeping it fully compatible with multi-map boost control. The multiplier applies on top of whichever map's wastegate target is currently active, so it works consistently across all three maps without any additional setup.





Changes: HMT Echo Engine Reconditioning Program

New LUA

New Lua has been written for the combustion engine power curve logic, adding camshaft and ported head gain factors integrations. This cleans up the math for the power gains allowing them to be independent from the intake and exhaust gain factors and scale cleanly with additional modifications within the formula.

HMT Ported Head

The ported head now operates as a scaling multiplier, contributing up to approximately 11–12% across the RPM range rather than a fixed delta. It also integrates improved exhaust flow into the turbo's power calculation, reflecting the real-world benefits of spool as a result of the greater exhaust gas volume and velocity that comes with ported. In short, head porting now rewards a complete build rather than delivering a flat number regardless of context.

Revised Engine Long Block System

The vanilla engine long block slot, previously a subslot of the displacement slot, has been retired. In its place, the gains that previously lived in the "Stage 1" and "Stage 2" long block selections have been redistributed across two new, independent entities: **Bottom End** and **Camshaft & Valvetrain**.

Stroker's

Patrons get access to a 2.7L and a 3.4L displacement variants, both built with lighter weight internals for improved top end capabilities!

Bottom End

The Bottom End slot governs internals: rev limit, peak torque ceiling, rotating assembly mass, internal friction, and compression ratio. The Stage 1 and Stage 2 bottom end additionally carry a portion of the power multiplier previously attributed to the full long block. This is a 7% gain across the rev range on Stage 1, and 11% on Stage 2. The remainder of those long block gains has been moved to the camshaft, as detailed below.

HMT Valvetrain

The prior "Echo Camshaft" has been renamed the **HMT Street Camshaft** and recalibrated. When paired with the Ported Head, it delivers approximately 13% additional peak power over the head alone. As a real-world reference point: on *Octavia*, a client build commissioned out of Magdeburg, the combination brought output from 435bhp & 576nm to 496bhp & 647nm

Two additional camshaft options are now available:

- **HMT Sport Camshaft**: Pairs with the Stage 1 Bottom End, contributing a 35% torque multiplier on top of the bottom end's gains.
- **HMT Competition Camshaft**: Pairs with the Stage 2 Bottom End, contributing a 50% torque multiplier on top of the bottom end's gains.

Cumulatively, the full stack matches the output ceiling of the old long block system, now just with each component selectable and upgradeable independently and the addition of the ported head multiplier.



NEW - HMT 'Atem' Individual Throttle Bodies:

Methodology

Individual throttle bodies are a cult modification for good reason. On paper, the numbers rarely justify them, as a turbocharger will always deliver more power per dollar and per install hour. But we all know that's never been the point.

What ITBs offer is something a turbocharger fundamentally cannot: instant response and sound. There's no waiting for boost, just linear and direct power delivery from idle upward. And of course, the intake noise, particularly from a straight-six's natural harmonic characteristics, is among the finest sounds any internal combustion engine produces.

HMT has offered NA intake packages before under the "Eden" name for the modern V8 found in the K-Series ttSport. The "Atem" program extends the Eden programs philosophy to the I-Series with six 45mm independent throttle bodies taking the place of the factory intake manifold allowing each cylinder **maximum air flow**.

Offering

The Atem ITBs are compatible with all displacement variants and internal modifications. It can be best paired with the Echo Standalone ECU with the "Maps For" slot set to "Naturally Aspirated.' They pair particularly well with the HMT 2.7L Stroker (Patreon), the Echo Ported Head, and the Street or Sport Camshaft. In that configuration, expect around 270whp at 6,200 RPM, although the number is beside the point, the experience should be the driving reason towards pursuing this setup.

Recommended Setups

If you are looking for build guidance, ***HMT*** recommends the following setups for optimal performance with the ITBs

Displacement	Internals	Head	Cams	BHP	Nm
2.4L	*Stage 2*	*Ported*	*Sport*	*242bhp*	*289nm*
3.0L	*Stage 1*	*Ported*	*Street*	*292bhp*	*353nm*



NEW - HMT Rolling ALS for ALL PLATFORMS BETA

Methodology

HMT's Rolling Anti-Lag System is a fully custom Lua implementation designed around a single principle: rolling ALS should only ever be active under full throttle with the driver switch engaged. This distinguishes it from conventional bypass-style anti-lag, which operates off-throttle, and is the same behavior found in BeamNG's vanilla ALS controller.

Earlier versions of this feature in game (*credit to Zeit for his version of this concept*) built rolling ALS off of the vanilla off-throttle model. ***HMT's*** approach was similar, with our implementation inverting the throttle requirement. The system is entirely inert unless:

1. the ALS switch (default key bind is 'alt+a') is held down,
2. the engine RPM is above 3500,
3. throttle is at WOT simultaneously.

Each condition on its own will not trigger it, **all** have to be true for activation.

Auto-disarm:

Continuous activation is capped at 5 seconds, after which the system disarms automatically to protect the turbocharger from sustained thermal stress.

BETA - Integration is ongoing. Overall functionality is stable and operating as intended but edge cases and UI implementation are still being refined.

Remaster – Pure430b Gen2 Remaster:

Methodology

Following an internal review of its performance in the market, ***HMT*** has revised the Pure430b package. Our Gen1 **kit underperformed commercially**. Unlike the rest of ***HMTs*** "Pure" lineup, it **required a built engine**, creating a **cost-prohibitive** barrier-to-entry that most owners of entry level ETKs **aren't willing to cross**. The 844/854 and Kc4s are designed as a cost effective yet premium, efficient, daily driver. The few owners who do modify them can **rarely justify the investment** required of the Gen1 kit with its built engine, when their money would go further with a 6 cylinder and less effort.

With all of this in mind, we began development of the **Pure430b Gen2**. The primary objective was **reducing peak torque** to bring the package **within stock long block limits**, eliminating the built engine requirement entirely. Through this, the goals became to **retain factory like drivability** with reasonably quick spool, but **bolster the top end power** and **eliminate** the characteristic **torque spike effect** of both the factory turbo and our Gen1 unit for longevity and refinement. We wanted to do all of this while still retaining, if not exceeding, the performance of the Gen1 package. Quite a task considering the amount of torque we would need to drop.

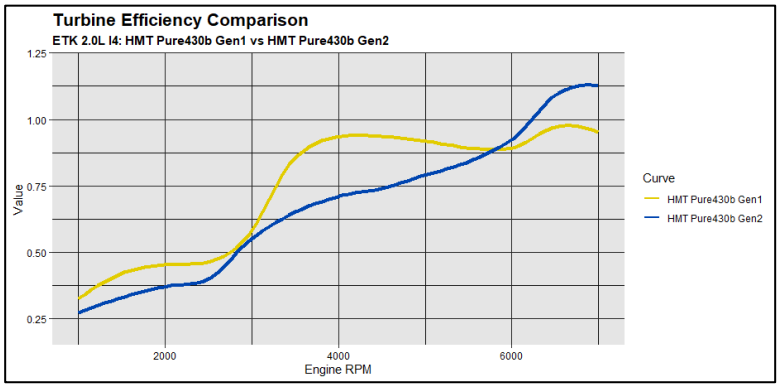
Changes

From a software standpoint, revised boost control and updated ignition timing maps **suppress the low to-mid RPM torque spike** that defined Gen 1's delivery.

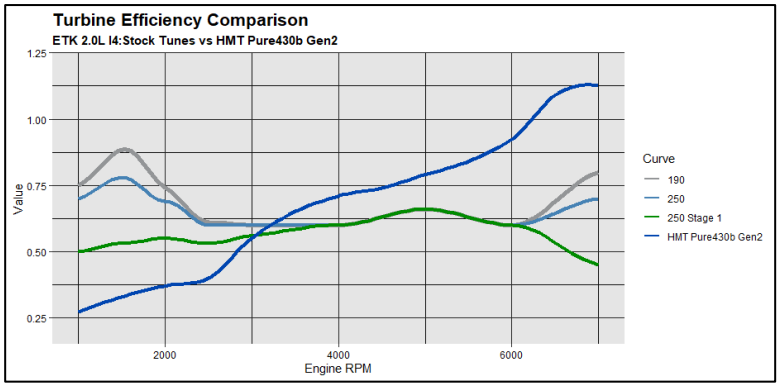
On the hardware side, the Gen2 turbo features a **smaller billet compressor wheel** and **reduced A/R turbine housing** relative to Gen 1, which both contribute to **earlier spool**, **better low-end torque** and **transient response**, and more **linear power delivery** all while sitting in the stock location.

Output

Peak power remains **430bhp** on 98 RON (93 AKI), with an **E85 map now available** pushing output to **460bhp** for those with ethanol available, mirroring our ecu system structure from the Pure560b. **Torque is down by design**, making 453nm (335 lb-ft) versus the prior 579nm (420 lb-ft). Those who shop by peak figures alone will see that as a step backwards, but the reality on the road is quite the opposite.



Power arrives earlier, **holds cleaner to redline**, and the Gen2 package stays **near even** with the prior Gen1 **everywhere it counts** since the drivetrain isn't trying to recover from a torque spike it was never really built to handle. On the ethanol map, the Gen2 Pure430b pulls away from the Gen1 easily, putting almost **2 cars lengths on it in the quarter despite being down almost 100nm of torque**. As we have always strived to prove, peak numbers are not the most important thing.





Remaster – HMT Q2 Boost Gauges (Patreon Feature)

Minor but still meaningful changes to the HMT Q2 Boost gauge!

Code

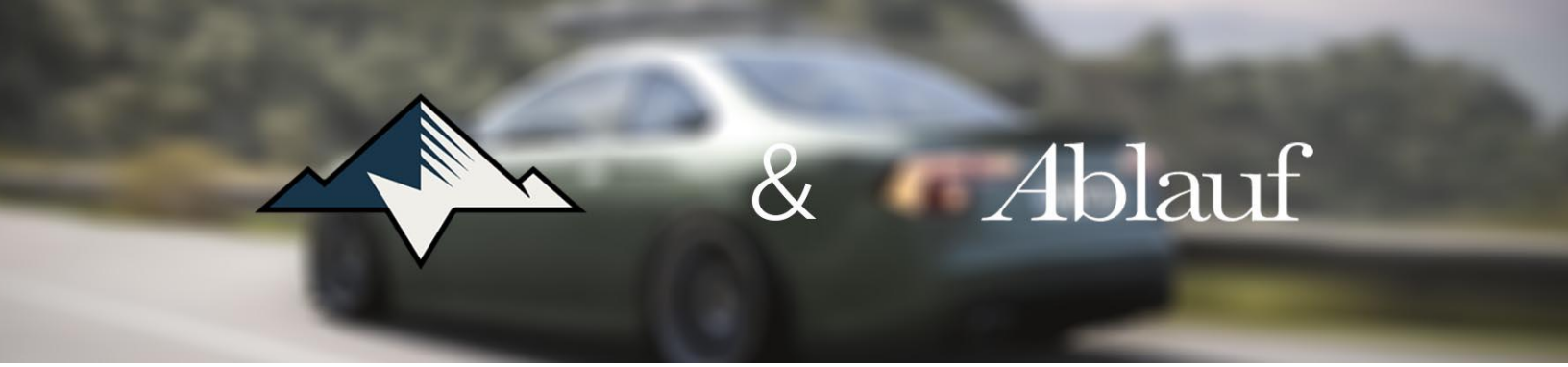
The gauge's underlying HTML/JS has been reworked for cleaner structure and improved runtime efficiency. No behavioral changes, just a leaner implementation under the hood. May rewrite to match BeamNG's new VUE format in the future to get it fully up to modern standards and eliminate buffering entirely.

LED Bar Colors

- Orange (Base): matches standard gauge cluster
- Blue (ttSport): matches ttSport gauge cluster
- Heritage Gold: special edition that matches the "heritage gold" accents on the Heritage cars
- White
- Red

Position:

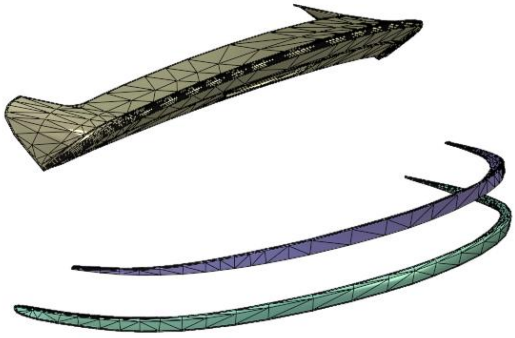
On the ETK 800, the gauge has been relocated from the bottom of the vent stack to the top. This improves vertical modularity in the HTML layout and puts the gauge in a more natural sightline.



NEW - HMT & Ablauf Aero Parts for Modern ETKs

Partnership

Harzer Mountain Tuning has contracted Ablauf in the manufacturing of aerodynamic components for modern ETK cars. Ablauf, a brand founded by @ghost_cauli, was established in 1991 in Yokohama, Japan. Their distinct stylistic catalog and the German roots of their brand name prompted ***HMT*** to reach out and establish a formal collaboration.



There are currently two product suites for modern ETKs produced by Ablauf, available exclusively through ***HMT***. The lineup consists of a **wagon trunk lip for the ETK 85X** and **four lip variants for the K-Series**.

The K-Series suite gets **complicated**, so here is how it works. The K-Series offers a standard and a sport trunk. The sport trunk features an integrated CSL-style lip, meaning both trunks carry different geometry and have **different available spoiler slots.**



The **standard trunk** exposes two slots: a dedicated **Lip Spoiler slot** and a **Spoiler slot**. The variants of the Ablauf Type A & B lips for the standard trunk are found in the Lip Spoiler slot. The Spoiler slot on the standard trunk is reserved for the larger vanilla aero parts.

The **sport trunk** exposes only the **Spoiler slot**. No Lip Spoiler slot is present, as the integrated lip geometry replaces it. As such, the sport variants of the Ablauf Type A & B lips are in the Spoiler slot with the rest of the larger vanilla aero parts



Quick reference for which variant for which trunk:

- Standard trunk → install via the **Lip Spoiler slot**
    - **Do not** install via the Spoiler slot
- Sport trunk → install via the **Spoiler slot**

Note: the lips for the sport trunk will populate in the Spoiler slot in the standard trunk too despite not being meant for it. This is an unavoidable limitation of the slot structure of the two parts. It will leave a large visible gap if setup this way, which obviously is not an intended look.